Day-36

Discovery Protocols (CDP & LLDP)

Layer 2 discovery protocols

- Layer 2 discovery protocols such as CDP and LLDP share information with and discover information about neighboring (connected devices)

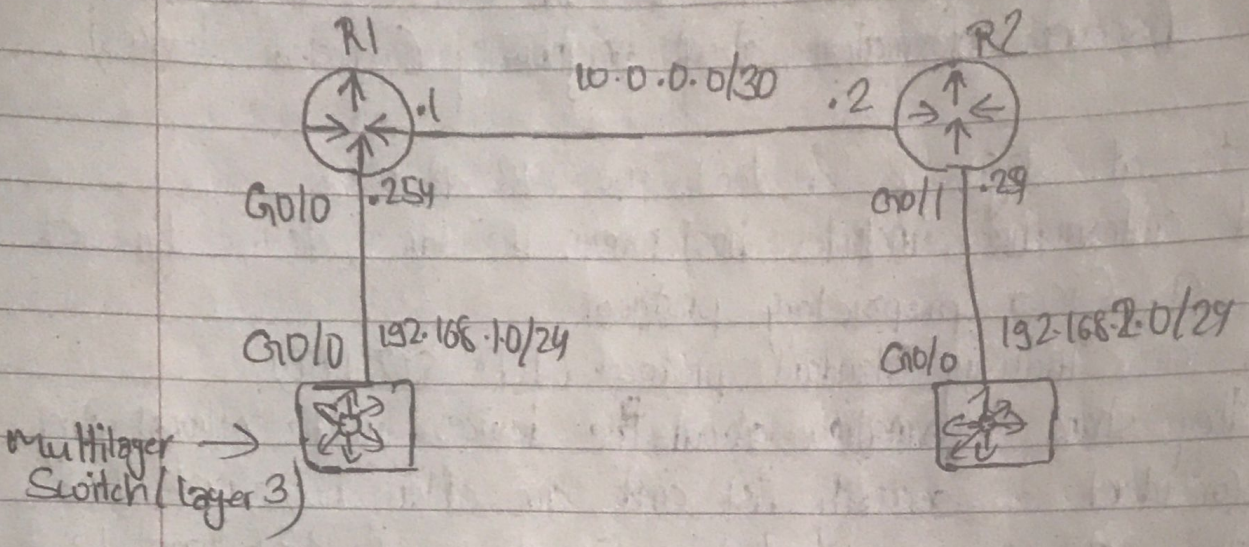
- They operate at layer 2 & don't use IP address
- The shared information includes host name, IP address, device type etc.
- CDP is a Cisco proprietary protocol.
- LLDP is an industry standard protocol (IEEE 802.1AB)
- Because they share information about the devices in the network, they can be considered a security risk and are often not used. It is upto the network engineer/admin to decide if they want to use them in network or not

Cisco Discovery Protocol (CDP)

- CDP is a Cisco proprietary protocol
- It is enabled on Cisco devices (routers, switches, firewalls, IP phones, etc) by default.
- CDP messages are periodically sent to MAC 0100.0ccc.cccc.
- When a device receives a CDP mssg, it processes and discards the message. It doesn't forward it to other device

- By default CDP messages are sent once every 60 seconds.
- By default, the CDP holdtime is 180 seconds. If a message isn't received from a neighbor for 180 seconds, the neighbor is removed from the CDP neighbor table.
- CDPv2 mssg are sent by default

CDP CLI



- R1 # show cdp.
  Global CDP info:
  CDP version
  Holdtime
  default time

  [Note: If CDP isn't enabled on device, it shows "% CDP is not enabled" on CLI]

- R1 # show cdp traffic
  Total packets output: input:
  Checksum, Encaps failed.
  v1 traffic
  v2 traffic.

- R1 # show cdp interface
  It gives info about all interfaces

- R1 # show cdp neighbors

device id	Local int	Holdtime	Capability	Platform	Port ID
hostname of R1 - CDP - naighbors	Interface on local device on R1	Each time R1 Receives CDP mssg it resets to 180	helps to identify what kinda device we're connected to	model of neighboring devices	Port id or int no on neighboring device

- R1 # show cdp neighbors detail
  More information is displayed for each neighbor like
  S Version of IOS:
  VTP information: { LLDP won't show VTP info as VTP is Cisco Proprietary but LLDP isn't }
  Native VLAN: { Helps to identify mismatch of native vlan
  Duplex: int duplex }

- R1 # show cdp entry R2 (ie sh cdp entry neighbor_host-name)
  It is same as above but only for a single neighbor device.

CDP configuration commands

- CDP is globally enabled by default.
- CDP is also enabled on each interface by default.
- To enable/disable CDP globally: R1(config)# [no] cdp run
- To enable/disable CDP on specific interfaces: R1(config-if)# [no] cdp enabl
- Configure the CDP timer: R1(config)# cdp timer seconds.
- Configure the CDP holdtime: R1(config)# cdp holdtime seconds.
- Configure Enable/disable CDPv2: R1(config)# [no] cdp advertise-v2

Scanned with CamScanner

6 Link Layer Discovery Protocol

- LLDP is an industry standard protocol (IEEE 802.1AB)
- It is usually disabled on Cisco devices by default, so it must be manually enabled.
- A device can run CDP & LLDP at the same time.
- LLDP messages are periodically sent to multicast MAC address 0180.C200.000E

- When a device receives an LLDP message, it processes and discards the message. It doesn't forward it to other devices

- By default, LLDP messages are sent once every 30 seconds
- By default, the LLDP holdtime is 120 seconds
- LLDP has an additional timer called the 'reinitialization delay'. If LLDP is enabled (globally or on an interface) this timer will delay the actual initialization of LLDP. 2 seconds by default.

LLDP IOS configuration commands

- LLDP is usually globally disabled by default
- LLDP is also disabled on each interface by default
- To enable LLDP globally.
  R1(config-if) # lldp run
- To enable LLDP on specific interfaces (tx).
  R1(config-if) # lldp transmit

Scanned with CamScanner

- To enable lldp on receive dirⁿ (rx):
  R1(config-if)# lldp receive.

- Configure the LLDP timer:
  R1(config)# lldp timer seconds

- show lldp
- show lldp traffic
- show lldp neighbors
- show lldp neighbors detail

(same as CDP